MSD MANUAL
Professional Version

Influenza

(Flu)

*By **Sophie Katz**, MD, MPH, Vanderbilt University Medical Center*
*Reviewed By **Brenda L. Tesini**, MD, University of Rochester School of Medicine and Dentistry*
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Influenza is a viral respiratory infection causing fever, coryza, cough, headache, and malaise. Mortality is possible during seasonal epidemics, particularly among high-risk patients (eg, those who are institutionalized, at the extremes of age, have cardiopulmonary insufficiency, or are in late pregnancy); during pandemics, even healthy, young patients may die. Diagnosis is usually clinical and depends on local epidemiologic patterns. Antiviral treatment reduces the duration of illness by about 1 day and should be specifically considered for high-risk or severely ill patients. The influenza vaccine should be given annually to all eligible patients who do not have a contraindication.

Influenza refers to illness caused by the influenza viruses, but the term is commonly and incorrectly used to refer to similar illnesses caused by other viral respiratory pathogens. Influenza viruses are classified as type A, B, or C by their nucleoproteins and matrix proteins. Influenza C virus infection does not cause typical influenza illness and is not discussed here.

Influenza antigens

Hemagglutinin (H) is a glycoprotein on the influenza viral surface that allows the virus to bind to cellular sialic acid and fuse with the host cell membrane. Neuraminidase (NA), another surface glycoprotein, enzymatically removes sialic acid, promoting viral release from the infected host cell. There are 18 H types and 11 NA types, giving 198 possible combinations, but only a few are human pathogens.

Antigenic drift refers to relatively minor, progressive mutations in preexisting combinations of H and NA antigens, resulting in the frequent emergence of new viral strains. These new strains may cause seasonal epidemics because protection by antibody generated to the previous strain is decreased.

Antigenic shift refers to the relatively rare development of new combinations of H and/or NA antigens, which result from reassortment of subunits of the viral genome. Pandemics can result from antigenic shift because antibodies against other strains (resulting from vaccination or natural infection) provide little or no protection against the new strain.

Epidemiology of Influenza

Influenza causes widespread sporadic illness yearly during fall and winter in temperate climates (seasonal epidemics).

Seasonal epidemics are caused by both influenza A and B viruses; since 1968, most seasonal influenza epidemics have been caused by H3N2 (an influenza A virus). Influenza B viruses may cause milder disease but often cause epidemics with moderate or severe disease, either as the predominant circulating virus or along with influenza A.

Most influenza epidemics are caused by a predominant serotype, but different influenza viruses may appear sequentially in one location or may appear simultaneously, with one virus predominating in one location and another virus predominating elsewhere.

A weekly surveillance report of seasonal influenza in the United States is available at the Centers for Disease Control and Prevention's FluView.

Pandemics are much less common. There have been 6 major influenza pandemics, typically named after the presumed location of origin:

- 1889: Russian influenza (H2N2)
- 1900: Old Hong Kong influenza (H3N8)
- 1918: Spanish influenza (H1N1)
- 1957: Asian influenza (H2N2)
- 1968: Hong Kong influenza (H3N2)
- 2009: Swine influenza (influenza A [H1N1]pdm09)

In 2009–2010 an H1N1 flu pandemic occurred—the virus spread to > 70 countries and throughout the United States (1). The majority of the deaths occurred in Mexico. The virus was initially referred to as a swine flu virus, but it is a combination of swine, avian, and human influenza viruses. The infection is not acquired through ingestion of pork and is acquired very rarely by contact with infected pigs. Subsequently, the virus name was standardized to influenza A(H1N1)pdm09 to denote the pandemic and distinguish the virus from seasonal H1N1 strains and the 1918 pandemic H1N1 strain. Since 2009, influenza A(H1N1)pdm09 has circulated as a seasonal influenza.

Influenza viruses can be spread by

- Airborne droplets
- Person-to-person contact
- Contact with contaminated items

Airborne spread appears to be the most important mechanism.

High-risk groups

Certain patients are at high risk of complications from influenza:

- Children < 5 years; children < 2 years are at particularly high risk
- Adults > 65 years
- People with chronic medical disorders (eg, cardiopulmonary disease, diabetes mellitus, renal or hepatic insufficiency, hemoglobinopathies, immuodeficiency)
- Women in the second or third trimester of pregnancy
- Patients with disorders that impair handling of respiratory secretions (eg, cognitive dysfunction, neuromuscular disorders, stroke, seizure disorders)
- Patients ≤ 18 years taking aspirin (because Reye syndrome is a risk)

Morbidity and mortality in these patients may be due to exacerbation of underlying illness, acute respiratory distress syndrome, primary influenza pneumonia, or secondary bacterial pneumonia.

Epidemiology reference

1. Sullivan SJ, Jacobson RM, Dowdle WR, Poland GA: 2009 H1N1 influenza. *Mayo Clin Proc*. 2010;85(1):64-76. doi:10.4065/mcp.2009.0588

Symptoms and Signs of Influenza

The incubation period for influenza ranges from 1 to 4 days with an average of 48 hours. In mild cases, many symptoms are like those of a common cold (eg, sore throat, rhinorrhea); mild conjunctivitis may also occur.

Typical influenza in adults is characterized by sudden onset of chills, fever, prostration, cough, and generalized aches and pains (especially in the back and legs). Headache is prominent, often with photophobia and retrobulbar aching. Respiratory symptoms may be mild at first, with scratchy sore throat, substernal burning, nonproductive cough, and sometimes coryza. Later, lower respiratory tract illness becomes dominant; cough can be persistent, raspy, and productive.

Gastrointestinal symptoms may occur and appear to be more common with the 2009 pandemic H1N1 strain (1). Children may have prominent nausea, vomiting, or abdominal pain, and infants may present with a sepsis-like syndrome.

After 2 to 3 days, acute symptoms rapidly subside, although fever may last up to 5 days. Cough, weakness, sweating, and fatigue may persist for several days or occasionally for weeks.

Complications

Pneumonia is suggested by a worsening cough, bloody sputum, dyspnea, and rales. Secondary bacterial pneumonia is suggested by persistence or recurrence of fever and cough after the primary illness appears to be resolving.

Encephalitis, myocarditis, and myoglobinuria, sometimes with renal failure, develop infrequently after influenza A or B infection. Reye syndrome—characterized by encephalopathy; fatty liver; elevation of

liver enzymes, ammonia, or both; hypoglycemia; and lipidemia—often occurs during epidemics of influenza B, particularly in children who have ingested aspirin.

Symptoms and signs reference

1. Cao B, Li XW, Mao Y, et al. Clinical features of the initial cases of 2009 pandemic influenza A (H1N1) virus infection in China. *N Engl J Med*. 2009;361(26):2507-2517. doi:10.1056/NEJMoa0906612

Diagnosis of Influenza

- History and physical examination
- Sometimes rapid antigen or reverse transcriptase-polymerase chain reaction (RT-PCR) tests
- Pulse oximetry and chest radiograph for patients with severe respiratory symptoms

The diagnosis of influenza is generally made clinically in patients with a typical syndrome when influenza is known to be present in the community.

Although many rapid diagnostic molecular and antigen detection tests are available and most have good specificity, their sensitivities vary widely, and they usually add little to patient management. (See Centers for Disease Control and Prevention (CDC): Overview of Influenza Testing Methods). Diagnostic tests should be done when results will affect clinical decisions.

RT-PCR tests are sensitive and specific and can differentiate influenza types and subtypes. If this test is quickly available, results may be used to select appropriate antiviral therapy; it should also be done when influenza is suspected in hospitalized patients because antiviral treatment is usually indicated (1). Also, these tests can prevent the unnecessary use of antibacterial medications, and identification of the specific influenza virus can be important for infection control. These tests are also useful to determine whether outbreaks of respiratory disease are due to influenza.

Cell culture of nasopharyngeal swabs or aspirates takes several days and is not useful for patient management decisions.

If patients have lower respiratory tract symptoms and signs (eg, dyspnea, rales noted during lung examination), pulse oximetry to detect hypoxemia and a chest radiograph to detect pneumonia should be done. Primary influenza pneumonia appears as focal or diffuse interstitial infiltrates or as acute respiratory distress syndrome. Secondary bacterial pneumonia is more likely to be lobar or segmental.

Diagnosis reference

1. Uyeki TM, Bernstein HH, Bradley JS, et al: Clinical Practice Guidelines by the Infectious Diseases Society of America: 2018 Update on Diagnosis, Treatment, Chemoprophylaxis, and Institutional Outbreak Management of Seasonal Influenza [published correction appears in *Clin Infect Dis*. 2019 May 2;68(10):1790]. *Clin Infect Dis*. 2019;68(6):e1-e47. doi:10.1093/cid/ciy866

Treatment of Influenza

- Symptomatic treatment
- Sometimes antiviral medications

Treatment for most patients with influenza is symptomatic; it includes rest, hydration, and antipyretics as needed, but aspirin is avoided in patients ≤ 18 years (due to risk of Reye syndrome). Complicating bacterial infections require appropriate antibiotics.

Medications for influenza

Antiviral medications given within 1 to 2 days of symptom onset decrease the duration of fever, severity of symptoms, and time to return to normal activity. Treatment with antiviral medications is recommended for severely ill patients and high-risk patients (including all hospitalized patients) who develop influenza-like symptoms; this recommendation is based on data suggesting that early treatment may prevent complications in these patients.

Medications for influenza include the following (see CDC: Influenza Antiviral Medications: Summary for Clinicians):

- Oseltamivir, zanamivir, and peramivir (neuraminidase inhibitors)
- Baloxavir marboxil (endonuclease inhibitor)

Neuraminidase inhibitors interfere with release of influenza virus from infected cells and thus halt spread of infection.

The endonuclease inhibitor baloxavir marboxil interferes with viral replication by blocking viral RNA transcription. It is active against influenza A and B and may be an important new treatment option should resistance to neuraminidase inhibitors develop.

Zanamivir is given by an inhaler; it can be used in adults and children ≥ 7 years. Zanamivir sometimes causes bronchospasm and should not be given to patients with reactive airway disease; some people cannot use the inhalation device.

Oseltamivir is given to patients > 12 years; lower doses may be used in children as young as 2 weeks of age. Oseltamivir may cause occasional nausea and vomiting. In children, oseltamivir may decrease the incidence of otitis media; however, no other data clearly show that treatment of influenza prevents complications.

Peramivir is given IV as a single dose and can be used in patients > 6 months who cannot tolerate oral or inhaled medications. Studies of its use for influenza B are limited.

Baloxavir marboxil is given as a single oral dose to patients ≥ 5 years with uncomplicated influenza who have been symptomatic for ≤ 48 hours and who are otherwise healthy or at high-risk of developing complications (1, 2).

Adamantanes (amantadine and rimantadine) were previously used; however, more than 99% of current and recent circulating influenza viruses are resistant to adamantanes, so these medications are currently not recommended for treatment. Adamantanes block the M2 ion channel and thus interfere with viral uncoating inside the cell. They were effective only against influenza A viruses (influenza B viruses lack the M2 protein).

Treatment references

1. Hayden FG, Sugaya N, Hirotsu N, et al: Baloxavir marboxil for uncomplicated influenza in adults and adolescents. *N Engl J Med* 379:913-923, 2018. doi:10.1056/NEJMoa1716197
2. Ison MG, Portsmouth S, Yoshida Y, et al: Early treatment with baloxavir marboxil in high-risk adolescent and adult outpatients with uncomplicated influenza (CAPSTONE-2): a randomised, placebo-controlled, phase 3 trial. *Lancet Infect Dis* 20(10):1204-1214, 2020. doi: 10.1016/S1473-3099(20)30004-9. Epub 2020 Jun 8. PMID: 32526195.

Prognosis for Influenza

Most patients recover fully, although full recovery often takes 1 to 2 weeks. However, influenza and influenza-related pneumonia are important causes of morbidity or mortality in high-risk patients. Prompt antiviral treatment in these patients can reduce the incidence of lower respiratory disease and hospitalization. Appropriate antibacterial therapy decreases the mortality rate due to secondary bacterial pneumonia.

Overall, the case fatality rate is low (eg, < 1%), but because incidence of disease is high, the total number of deaths can be significant. The Centers for Disease Control and Prevention (CDC) estimates that in the United States from 2010 to 2023, hospitalizations resulting from seasonal influenza ranged from 100,000 to 710,000 annually, and deaths ranged from 4,900 to 51,000 annually (1). Rates of hospitalization and death are highest in patients > 65 years. During typical seasonal influenza epidemics, approximately 80% of deaths are estimated to occur in patients > 65 years; however, 80% of H1N1-related deaths were estimated to have occurred in people < 65 years during the first 12 months of the 2009 H1N1 pandemic. (2, 3).

Prognosis references

1. CDC: Disease Burden of Flu. Accessed April 2024.
2. Dawood FS, Iuliano AD, Reed C, et al: Estimated global mortality associated with the first 12 months of 2009 pandemic influenza A H1N1 virus circulation: A modelling study. *Lancet Infect Dis*12 (9):687–695, 2012. doi: 10.1016/S1473-3099(12)70121-4
3. CDC:2009 H1N1 Pandemic (H1N1pdm09 virus). Accessed April 2024.

Prevention of Influenza

Influenza infections can largely be prevented by

- Annual vaccination
- Sometimes chemoprophylaxis (ie, with antiviral medications)

Influenza vaccines

For a detailed discussion of influenza vaccines, see Influenza Vaccine. Vaccination is indicated for all eligible patients but is especially important for high-risk patients and health care professionals.

Based on recommendations by the World Health Organization and the Centers for Disease Control and Prevention, influenza vaccines are modified annually to include the most prevalent strains (usually 2 strains of influenza A and 1 or 2 strains of influenza B). Sometimes slightly different vaccines are used in the northern and southern hemisphere.

Current commercially available influenza vaccines protect against seasonal H3N2, pandemic H1N1 influenza A, and influenza B. A complete list of influenza vaccines for the current season is available from the Centers for Disease Control and Prevention.

A vaccine for H5N1 avian influenza has been approved for people > 18 years at high risk of H5N1 exposure but is available only through public health officials. No vaccines are currently available for the other avian influenza viruses rarely associated with human disease (H7N7, H9N2, H7N3, and H7N9).

Antiviral medications

Although vaccination is the preferred method of prevention, antiviral medications are also effective.

Preexposure prophylactic antiviral medications can be considered during an epidemic for patients

- Who have been vaccinated only within the previous 2 weeks
- For whom vaccination is contraindicated
- Who are immunocompromised and thus may not respond to vaccination

Antiviral medications do not impair development of immunity from inactivated vaccine. They can be stopped 2 weeks after vaccination. If vaccine cannot be given, antiviral medications are continued for the duration of the epidemic.

Postexposure prophylactic antiviral medications are typically indicated for potentially exposed people when clusters of cases occur in a closed environment (eg, nursing home, hospital unit). These medications may also be given to household contacts or other exposed people at high risk of developing complications of influenza. Resistance patterns may affect medication choice.

Key Points

- **Minor antigenic drift in H and/or NA antigens produces strains that cause seasonal epidemics; rare antigenic shifts resulting in new combinations of H and NA antigens can cause a pandemic with significant mortality.**

- Influenza itself may cause pneumonia, or patients with influenza may develop secondary bacterial pneumonia.
- Diagnosis is usually clinical, but sensitve and specific RT-PCR assays can differentiate influenza types and subtypes and thus help select antiviral therapy and determine whether outbreaks of respiratory disease are due to influenza.
- Treat most patients symptomatically.
- Antiviral medications given early can slightly decrease duration and severity of symptoms but are typically used only in high-risk patients; different influenza types and subtypes are resistant to different medications.
- Vaccination is indicated for all eligible patients; antiviral medications can be used for prevention in immunocompromised patients (who may not respond to vaccination) and patients with contraindications to vaccination.

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